Kai Tom

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Qualifications

- Programs: Rhino 3D(11 years), Onshape(3), Solidworks(3), AutoCad(1), NX(1), Matlab(3), Python(2) Ansys FEA(1)
- Manufacturing Skills: 3D printers, Laser Printers, CNCs, Composite Layups, general shop tools
- Project Management: Systems Engineering, Workflow optimization, Informed decision making

Experience

Mechanical Engineer Intern | Panasonic Avionics - Irvine CA **June 2025 - September 2025**

- Designed and manufactured 34” CFRP curved monitor prototype for first class in-flight entertainment systems which reduced weight by 40% compared to existing aluminum / plastic unibody designs
- Scheduled and conducted weekly meetings with external manufacturers to perform DFM reviews using engineering drawings
- Designed parts to pass DO-160 (Commercial Aircraft Interior Abuse Loads Testing) testing standards
- Iterated through multiple design reviews and solicited team feedback to achieve a 15% reduction in panel thickness

Structures Lead | Design Build Fly at the UW - Seattle WA **May 2024 - Present**

- Simulated CFRP sandwich panel monocoque structures using composite FEA then validated with sub component testing to reduce overall aircraft weight by 35%
- Fabricated carbon fiber tooling molds ensure dimensional accuracy of 1.5mm per 1000mm
- Led a 15 person team in technical design reviews to make critical design decisions, while acting as the integration focal point, resulting in simplified load paths
- Mentored junior members and improved the team's technical capabilities, while simultaneously meeting project timelines and goals

Fuselage Subteam Lead | Design Build Fly at the UW - Seattle WA **September 2023 - June 2024**

- Led the design of the 5ft long carbon fiber skin-supported fuselage, custom parametric CAD system for rapid iteration and surface modeling for drag optimized external geometry
- Documented and justified design processes though hand calcs and FEA for a 60-page design report
- Facilitated DFM reviews to optimize assemblies for manufacturing and design manufacturing jigs, resulting in a decrease in wing construction time from 7 days to 2 days (350% reduction)
- Helped the team placed 3rd globally by providing critical tuning and repairs during the competition

Mechanical Specialist | Club Boring (Not Boring Company related) - Seattle **March 2023 - January 2025**

- Started a team to design and develop a remote operated solar powered ground vehicle
- Transferred domain specific knowledge to other team members as a subject matter specialist on mechanical design

Independent Engineering Projects

- **Solar Powered RC Plane -** Focused on optimizing structures and maximizing aerodynamic efficiency though experimental flight testing and optimizer scripts
- **Custom Hovercraft –** Designed and built a functional prototype with custom space frame and variable engine geometry
- **3D-Printed Speaker -** Featured a tuned back loaded horn and 30 minute final assembly time

Education

University of Washington | B.S. Mechanical Engineering **Expected Graduation: June 2026**

- Coursework (Through Spring 2025): Linkage Design, Dynamic Systems Analysis, Machine Design, Manufacturing Processes, Fluids, Heat Transfer, Thermodynamics, Mechanics of Materials
- Accolades - 3.84 GPA, Dean's List: Winter 2023 - Present